

SESSION-1

Task 1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Make file.

```
saichaitanya@GeethzzSai: ~/s × + v
saichaitanya@GeethzzSai:~$ cd skill_week1
saichaitanya@GeethzzSai:~/skill_week1$ nano main.c
saichaitanya@GeethzzSai:~/skill_week1$ |
```

CODE:

```
GNU nano 7.2 main.c
#include <stdio.h>

int main() {
    printf("Hello, OSSP Skill Week 1!\n");
    printf("GCC compilation is working successfully.\n");

    return 0;
}
```

OUTPUT:

```
saichaitanya@GeethzzSai:~$ cd skill_week1
saichaitanya@GeethzzSai:~/skill_week1$ nano main.c
saichaitanya@GeethzzSai:~/skill_week1$ gcc main.c -o main
saichaitanya@GeethzzSai:~/skill_week1$ ./main
Hello, OSSP Skill Week 1!
GCC compilation is working successfully.
saichaitanya@GeethzzSai:~/skill_week1$
```

```
GNU nano 7.2
CC = gcc
CFLAGS = -Wall -Wextra

all: main

main: main.c
    $(CC) $(CFLAGS) main.c -o main

clean:
    rm -f main|
```

Verifying linux environment by uname-a:

```
my Shell Project
saichaitanya@GeethzzSai:~/skill_week1$ uname -a
Linux GeethzzSai 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Th
u Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
saichaitanya@GeethzzSai:~/skill_week1$ |
```

Checking GCC C compiler:

```
saichaitanya@GeethzzSai:~/skill_week1$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
saichaitanya@GeethzzSai:~/skill_week1$ |
```

Git version:

```
saichaitanya@GeethzzSai:~/skill_week1$ git --version
git version 2.53.0
saichaitanya@GeethzzSai:~/skill_week1$ |
```

Task 2: To Analyse process abstraction, execute fork(), understand exec() family, analyse parent-child relationships, inspect process tree, practice system call tracing.

```
saichaitanya@GeethzzSai: ~/skill_week1$ nano task2.c
saichaitanya@GeethzzSai: ~/skill_week1$ |
```

CODE:

```
GNU nano 8.7.1 task2.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("Executing ls using exec()\n");
```

```
        execlp("ls", "ls", "-l", NULL);

        perror("exec failed");
        exit(1);
    }
    else {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent waiting for child...\n");

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

OUTPUT:

```
saichaitanya@GeethzzSai:~/skill_week1$ nano task2.c
saichaitanya@GeethzzSai:~/skill_week1$ gcc task2.c -o task2
saichaitanya@GeethzzSai:~/skill_week1$ ./task2
Parent Process Started
Parent PID: 1173

Parent Process
Parent PID: 1173
Child PID: 1174
Parent waiting for child...

Child Process
Child PID: 1174
Parent PID: 1173
Executing ls using exec()
total 116
-rwxr-xr-x 1 saichaitanya saichaitanya 16344 Aug 14 13:37 3
-rw-r--r-- 1 saichaitanya saichaitanya 636 Aug 14 13:37 3.c
-rwxr-xr-x 1 saichaitanya saichaitanya 16352 Aug 14 13:44 4
-rw-r--r-- 1 saichaitanya saichaitanya 1505 Aug 14 13:43 4.c
-rw-r--r-- 1 saichaitanya saichaitanya 131 Aug 14 13:18 Makefile
-rw-r--r-- 1 saichaitanya saichaitanya 0 Aug 14 13:13 README.md
drwxr-xr-x 2 saichaitanya saichaitanya 4096 Aug 14 13:13 include
-rwxr-xr-x 1 saichaitanya saichaitanya 15952 Aug 15 06:31 main
-rw-r--r-- 1 saichaitanya saichaitanya 151 Aug 15 06:31 main.c
drwxr-xr-x 2 saichaitanya saichaitanya 4096 Aug 14 13:13 src
-rwxr-xr-x 1 saichaitanya saichaitanya 16288 Aug 15 06:33 task2
-rw-r--r-- 1 saichaitanya saichaitanya 879 Aug 15 06:32 task2.c
-rwxr-xr-x 1 saichaitanya saichaitanya 16080 Aug 15 06:22 task3
-rw-r--r-- 1 saichaitanya saichaitanya 188 Aug 14 13:29 task3.c
drwxr-xr-x 2 saichaitanya saichaitanya 4096 Aug 14 13:13 tests
Child process completed.
saichaitanya@GeethzzSai:~/skill_week1$
```

SESSION-2

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

A) Create a file by using

```
saichaitanya@GeethzzSai:~/skill_week1$
saichaitanya@GeethzzSai:~/skill_week1$
saichaitanya@GeethzzSai:~/skill_week1$
saichaitanya@GeethzzSai:~/skill_week1$ nano 3.c
```

Then press enter and the code terminal will open then write a code

```

GNU nano 7.2
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myShell> ");
        fgets(input, sizeof(input), stdin);

        // Remove newline
        input[strcspn(input, "\n")] = 0;

        // Exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting Shell...\n");
            break;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}

```

After writing the code save the your program by ctrl O and for coming back to command press ctrl X then compile it by the command

```

saichaitanya@GeethzzSai:~/skill_week1$
saichaitanya@GeethzzSai:~/skill_week1$ nano 3.c
saichaitanya@GeethzzSai:~/skill_week1$ gcc 3.c -o 3
saichaitanya@GeethzzSai:~/skill_week1$ ./3

```

After compiling to get the output enter command

```

saichaitanya@GeethzzSai:~/skill_week1$ nano 3.c
saichaitanya@GeethzzSai:~/skill_week1$ gcc 3.c -o 3
saichaitanya@GeethzzSai:~/skill_week1$ ./3
myShell> ls
You entered: ls
myShell> OSp
You entered: OSp
myShell> exit
Exiting Shell...
saichaitanya@GeethzzSai:~/skill_week1$ |

```

Task 2: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

```

saichaitanya@GeethzzSai:~/skill_week1$ nano 4.c
saichaitanya@GeethzzSai:~/skill_week1$ gcc 4.c -o 4
saichaitanya@GeethzzSai:~/skill_week1$ ./4

```

CODE:

```
GNU nano 7.2
#include <stdio.h>

int main() {
    char buffer[100];
    int ch;
    int index = 0;

    printf("Type a command and press Enter:\n");
    printf("input> ");

    while ((ch = getchar()) != '\n' && ch != EOF) {
        if (ch == 127 || ch == 8) {
            if (index > 0) {
                index--;
            }
        } else if (index < 99) {
            buffer[index++] = ch;
        }
    }

    buffer[index] = '\0';

    printf("\nCommand entered: %s\n", buffer);
    printf("Characters stored: %d\n", index);

    return 0;
}
```

```
saichaitanya@GeethzzSai:~/skill_week1$
saichaitanya@GeethzzSai:~/skill_week1$
saichaitanya@GeethzzSai:~/skill_week1$
saichaitanya@GeethzzSai:~/skill_week1$ nano 4.c
saichaitanya@GeethzzSai:~/skill_week1$ gcc 4.c -o 4
saichaitanya@GeethzzSai:~/skill_week1$ ./4
Type a command and press Enter:
input> hello world

Command entered: hello world
Characters stored: 11
saichaitanya@GeethzzSai:~/skill_week1$ |
```